CLINICAL · EAR

Ear and mastoid operations: terms and principles

Ear operations are easiest to learn by what each one removes and what it keeps. The order of priorities never changes: a safe ear first, a dry ear second, hearing third. This page explains the terms and principles; the step-by-step guides are in Theatre.

AT A GLANCE

- Know each operation by what it removes and keeps.
- Cortical keeps the wall; modified radical keeps the hearing parts; radical removes them.
- Myringoplasty repairs the drum; tympanoplasty rebuilds hearing.
- Safety before hearing.
- Austin–Kartush classification (0, A–F) and MERI predict ossiculoplasty results.

1 CHAPTER ONE Overview

The language of ear surgery.

1.1 What hearing needs

For the middle ear to work, it needs an intact drum, a mobile ossicular chain, two mobile windows with *acoustic separation* (sound must reach the oval window but not the round window at the same time), an aerated middle ear with a working Eustachian tube, and a working cochlea. Every reconstruction tries to restore as many of these as possible.

1.2 Approaches

- **Transcanal (Rosen):** through the canal with a speculum, for stapes surgery, small perforations and exploratory tympanotomy.
- **Endaural (Lempert):** incisions in the canal extended up through the *incisura terminalis*, where there is no cartilage.

- **Postaural (Wilde):** behind the ear, for mastoid surgery and anterior perforations. In children under two, the mastoid tip is absent and the facial nerve is superficial, so the incision is shorter and more horizontal.

CHAPTER TWO

2 Myringotomy and grommets

Draining and ventilating.

2.1 Indications

Acute otitis media with a bulging, painful drum or complications; incomplete resolution; recurrent acute otitis media; otitis media with effusion; barotrauma; atelectasis; and access for intratympanic treatment.

2.2 Site

Posteroinferior for draining acute pus; anteroinferior or inferior for effusion and tubes. An anterosuperior tube stays in longest.

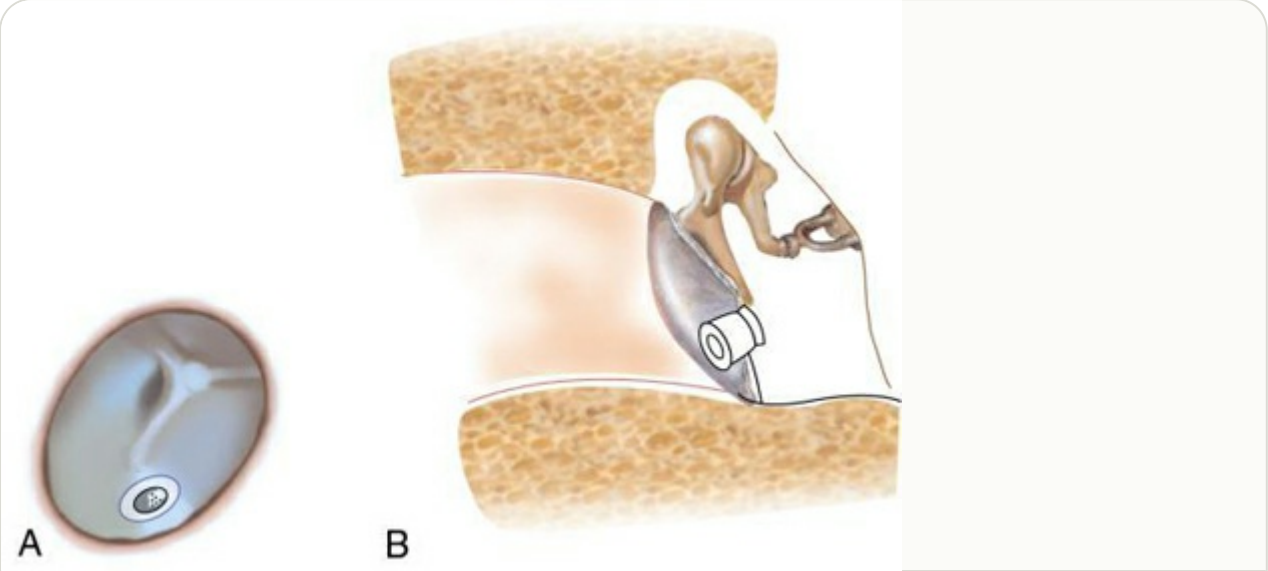


FIGURE 1 Grommets in the tympanic membrane.
Dhingra, fig. 10.4

2.3 Complications

Injury to the ossicles or a high jugular bulb, persistent perforation, tube blockage or extrusion, otorrhoea, granuloma and tympanosclerosis.

> **PEARL**
> Look for a vascular anomaly (a high jugular bulb or aberrant carotid) before cutting the drum.

3 CHAPTER THREE Repairing the drum and chain

Myringoplasty and tympanoplasty.

3.1 Myringoplasty

Repair of a perforation alone, without inspecting or reconstructing the middle ear. Dhingra distinguishes it from a type I tympanoplasty, in which the middle ear is examined.

3.1.1 Grafts

Temporalis fascia is the commonest; tragal perichondrium, cartilage (resists retraction) and vein are alternatives.

3.1.2 Techniques

- **Underlay:** the graft goes medial to the drum remnant and handle of malleus. Risks: a narrow middle ear space and anterior reperforation.
- **Overlay:** the graft goes lateral to the fibrous layer after removing the epithelium. Risks: blunting of the anterior angle, epithelial pearls and lateralisation of the graft.

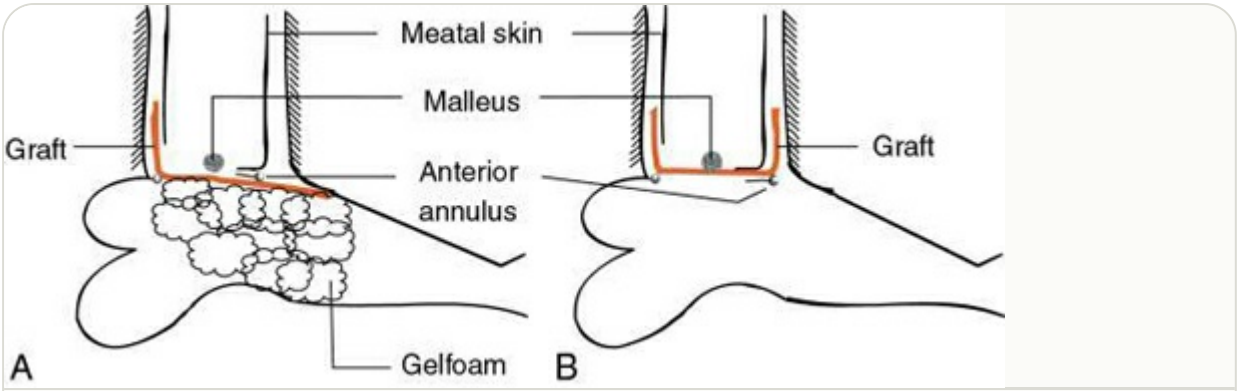


FIGURE 2 Myringoplasty: (A) the underlay technique, with the graft medial to the annulus and supported by gelfoam; (B) the overlay technique.

Dhingra, fig. 5.3

3.1.3 Small holes

A paper patch after cauterising the rim, or a fat plug.

3.1.4 Contraindications

Active discharge, uncontrolled nasal allergy, otitis externa, squamous epithelium growing into the middle ear, an only hearing ear, and young children under about three.

3.2 Tympanoplasty

Eradication of middle ear disease and reconstruction of the hearing mechanism, with or without mastoidectomy.

3.2.1 Wullstein's types

- **Type I:** graft on the handle of malleus, chain intact.
- **Type II:** graft on the incus, malleus missing.
- **Type III:** graft on the stapes head (myringostapediopexy).
- **Type IV:** only a mobile footplate; the graft shields the round window, creating a small air pocket.
- **Type V:** fixed footplate; a window is made in the lateral canal (Va) or the footplate removed (Vb).

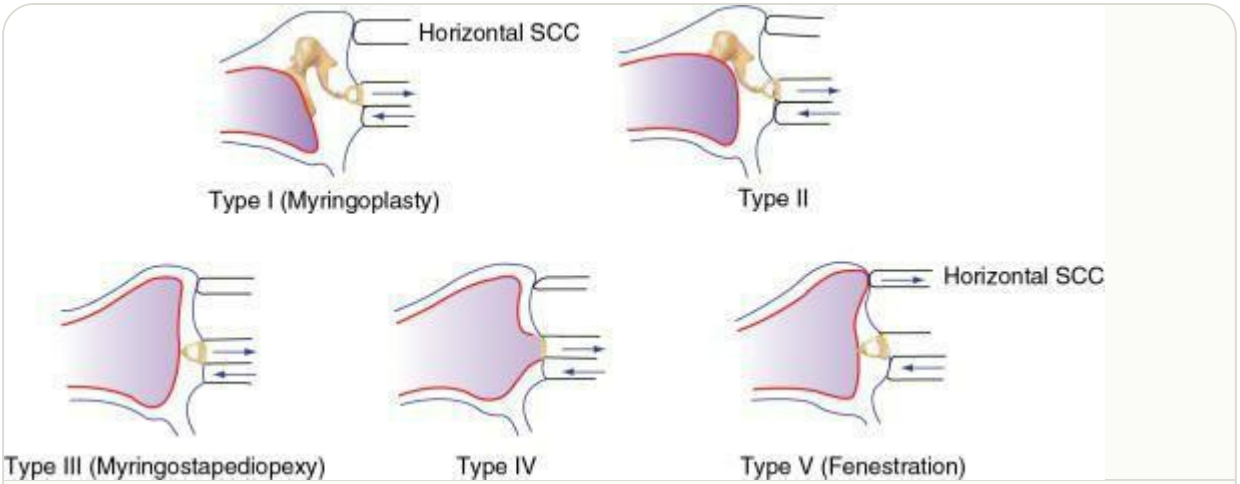


FIGURE 3 Types of tympanoplasty: the graft rests progressively on the malleus, incus, stapes and footplate.

Dhingra, fig. 5.2

3.2.2 Ossicular reconstruction

Reshaped autograft incus, or prostheses: a *PORP* (partial) sits on the stapes head; a *TORP* (total) on the footplate. Hydroxyapatite and titanium are common materials; a cartilage cap prevents extrusion. A PORP gives better hearing than a TORP.

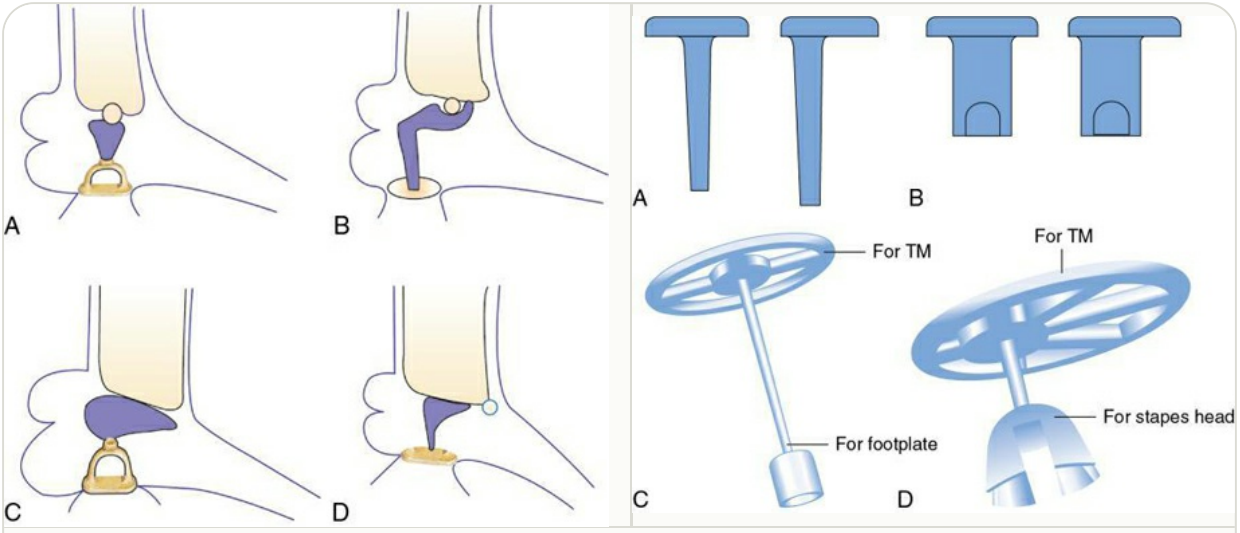


FIGURE 4 First, sculptured autograft ossicles. Second, hydroxyapatite and titanium partial and total prostheses.

Dhingra, figs 5.4 and 5.5

3.2.3 Timing

> Reconstruction is often delayed about six months after canal wall up surgery for cholesteatoma, to coincide with the second look; it is done primarily after trauma or for fixation.

4 CHAPTER FOUR Mastoid operations

What they remove.

4.1 Landmarks

MacEwen's (suprameatal) triangle marks the antrum, about 12 to 15 mm deep in adults. *Körner's septum*, a persistent petrosquamous plate, can be mistaken for the floor of the antrum. Other landmarks: the lateral semicircular canal, sinus plate, tegmen, digastric ridge and facial nerve.

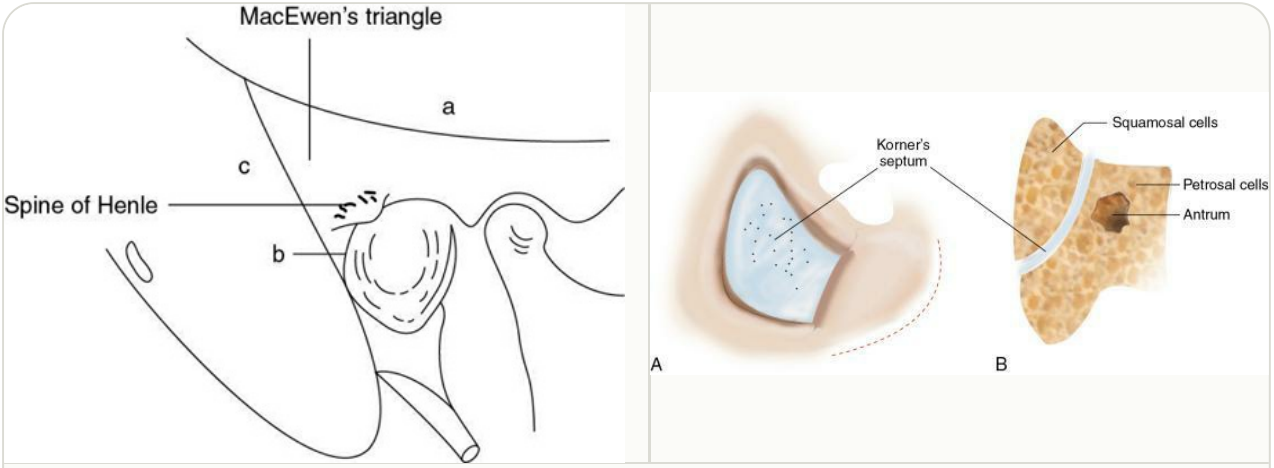


FIGURE 5 First, MacEwen's triangle. Second, Körner's septum, which may hide the antrum.
Dhingra, figs 1.12 and 1.14



FIGURE 6 MacEwen's triangle, and the middle ear seen through a posterior tympanotomy.
Anatomy atlas · ENT Steps p. 56, ENT Steps p. 46

4.2 Cortical (Schwartze) mastoidectomy

Removes the mastoid air cells and opens the antrum; the posterior canal wall and middle ear are left alone. For coalescent or masked mastoiditis, and as access for endolymphatic sac surgery, facial nerve decompression, cochlear implants and translabyrinthine surgery.

4.3 Modified radical mastoidectomy

The posterior canal wall is lowered and the attic, antrum and mastoid are exteriorised into one cavity, *keeping* the drum remnant, the working ossicles and the Eustachian tube. For cholesteatoma of the attic and antrum.

4.4 Radical mastoidectomy

Everything is exteriorised: the drum, malleus and incus and the middle ear lining are removed, and the Eustachian tube is plugged. For disease that cannot be cleared safely otherwise, failed surgery, and some tumours. Hearing loss is 50 dB or more.

MASTOID OPERATIONS COMPARED

OPERATION	POSTERIOR CANAL WALL	OSSICLES	HEARING
Cortical	Kept	Kept	Usually unchanged
Modified radical	Removed	Working parts kept	Can be reconstructed

OPERATION	POSTERIOR CANAL WALL	OSSICLES	HEARING
Radical	Removed	All but the stapes removed	Loss of 50 dB or more

4.5 Canal wall up and canal wall down

See the cholesteatoma topic for the choice between keeping and removing the posterior wall.

4.6 Meatoplasty

Every canal wall down cavity needs a wide meatus for ventilation and cleaning.

4.7 Obliteration

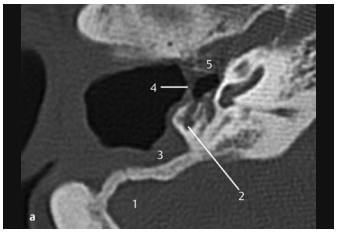
Filling a cavity with pedicled muscle, bone pâté or other material reduces discharge and care; disease must never be buried under it.

4.8 Complications of mastoid surgery

Facial nerve injury, lateral canal fistula with vertigo and hearing loss, dural injury and CSF leak, sigmoid sinus bleeding, and incus dislocation. About a quarter of radical cavities continue to discharge.

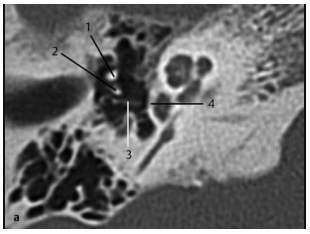
IMAGING IN THE APP 4 cases

A discharging cavity, piston position after stapedectomy, iatrogenic fenestration, and dural prolapse with CSF leak.



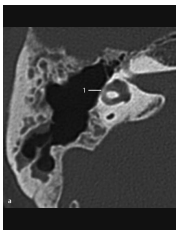
Modified Radical Cavity, Persistent Infections

After surgery · 2 images



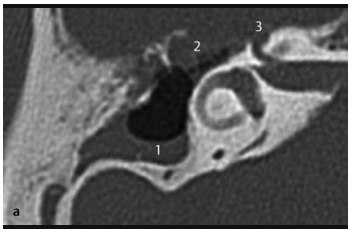
Stapedectomy, Control of Piston Position

After surgery · 4 images



Iatrogenic Fenestration

After surgery · 2 images



Dural Prolapse and Cerebrospinal Fluid Leakage

After surgery · 2 images

5 CHAPTER FIVE Before and after

Practical points.

5.1 Before

An audiogram and tuning forks before any ear operation; CT for cholesteatoma, revisions and suspected complications; check the other ear and Eustachian tube function.

5.2 After

Keep water out of the ear while a tube or graft is healing; review repaired ears at about three and six weeks; the graft is fully epithelialised in six to eight weeks. Clean mastoid cavities every four to six months in the first year, then yearly.

OPERATIVE GUIDES IN THEATRE 5 operations

Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation
Tubes

Myringoplasty and Tympanoplasty
Tympanoplasty

Ossicular Chain Reconstruction
Ossicular reconstruction

Mastoid Surgery
Mastoid surgery

Otosclerosis
Stapes surgery

OSSICULOPLASTY: CLASSIFICATION AND OUTCOMES (OXFORD TEXTBOOK)

The commonest defect is erosion of the long process of the incus and the incudostapedial joint, from chronic otitis media or retraction; trauma most often damages the incus. An air–bone gap over about 30 dB is a good indication for reconstruction, although smaller gaps are accepted when surgery is needed anyway for infection or cholesteatoma.

The **Austin–Kartush** classification describes what remains: **0** all ossicles present; **A** malleus and stapes present (incus eroded); **B** malleus present, stapes superstructure absent; **C** malleus absent, stapes present; **D** both absent; **E** fixed ossicular head; **F** fixed stapes. The **Middle Ear Risk Index (MERI)** adds scores for otorrhoea, perforation, cholesteatoma, ossicular status, granulation or effusion, previous surgery and smoking to predict the result.

Reconstruction depends on the stapes. With the stapes present, options are incus interposition or a reshaped ossicle, a **partial ossicular replacement prosthesis (PORP)**, bone cement (hydroxyapatite or glass ionomer) bridging a small incus defect, or myringostapediopexy. With only the footplate, a **total ossicular replacement prosthesis (TORP)** or autologous columella is used, sometimes with a footplate shoe for stability. Modern prostheses are titanium or hydroxyapatite and are covered by a cartilage plate to prevent extrusion. An air–bone gap under 20 dB is reached in about 60 to 90% with a stapes and 30 to 80% without. An active middle ear implant such as the Vibrant Soundbridge is an alternative when reconstruction repeatedly fails.

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Myringoplasty versus tympanoplasty.**
- **Wullstein's five types.**
- **Underlay versus overlay.**
- **Cortical, modified radical and radical mastoidectomy:** what each removes and keeps.

- **Mastoid landmarks:** MacEwen's triangle and Körner's septum.
- **Indications for myringotomy** and the choice of site.

6.2 Common pitfalls

COMMON PITFALLS

- Repairing a drum while the ear is still discharging.
- Mistaking Körner's septum for the floor of the antrum.
- Burying cholesteatoma under an obliteration.

Written for this app from Oxford Textbook of Otolaryngology ch. 19, 20, 21, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 20, Dhingra ch. 5, 78, 79, 80, 81, 82, 83 and K. J. Lee ch. 19. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

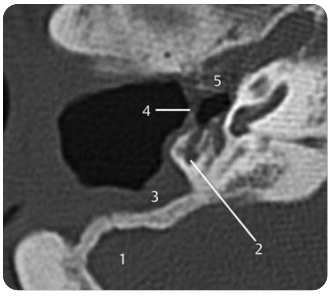
On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · AFTER SURGERY

Modified Radical Cavity, Persistent Infections

- Attention must be paid to the above-mentioned presence of residual cells. Residual cholesteatoma may present as smooth-bordered, rounded swellings covered by an epithelial or mucosal lining. In cases of destruction of the bony outline to the middle of posterior cranial fossa, a complementary MRI might be helpful for evaluating the extent of the pathology and its relation to intracranial structures.
- DWI is especially helpful in discriminating between recurrent or residual cholesteatoma and granulation tissue.



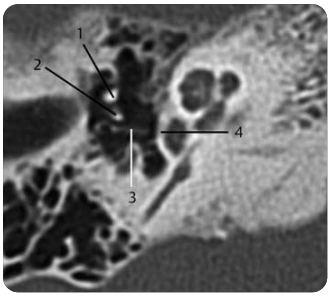
Persistent secretion from a modified radical cavity.

Dunnebier, Imaging for Otolaryngologists, Fig. 3.25 a

MIDDLE EAR & MASTOID · AFTER SURGERY

Stapedectomy, Control of Piston Position

- Ossicular prostheses may differ as regards their degree of radiopacity and visibility on CT. For this reason fine slices are required.
- Malpositioning of the piston around the incus and possible erosions or resorbtion of the long process and lenticular process due to compression of the piston are difficult to evaluate on CT.

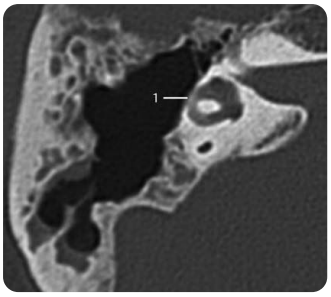


CT, axial. This view shows the handle of malleus (1) and posterior to it the long process of the incus (2), and a completely luxated pis- — ton (3) fro…

MIDDLE EAR & MASTOID · AFTER SURGERY

Iatrogenic Fenestration

- In fenestrated patients, no actual compression due to masses will be seen.
- Beware of suction of the cavity for cleaning. This must be performed very carefully to prevent severe vertigo and inner ear dysfunction.

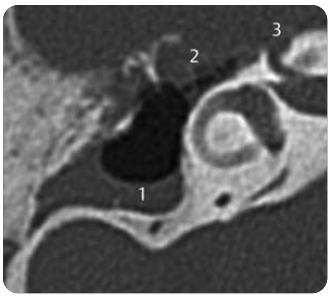


CT, axial. Before the availability of microsurgical techniques and the use of ossicular replacement prostheses, mobilization of the footplate was…

MIDDLE EAR & MASTOID · AFTER SURGERY

Dural Prolapse and Cerebrospinal Fluid Leakage

- Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement.



Cerebrospinal fluid leakage after insertion of a tympanostomy tube (grommet), indicated for middle ear secretions and conductive hearing…